

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot engine oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

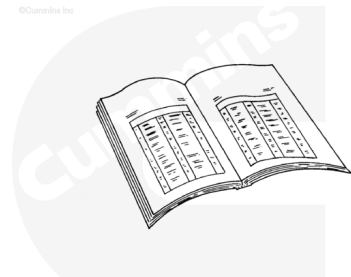
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

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ck800wa

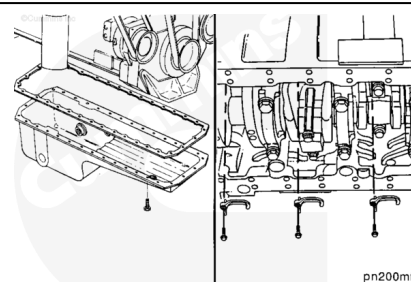
Note : Cummins Inc. does not recommend removing the cylinder liners to reduce oil consumption unless the cylinder liners are damaged and must be replaced.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html>)

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)

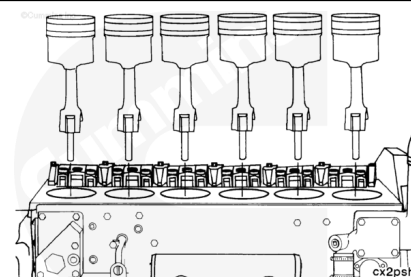
Initial Check

Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-046-tr.html>)



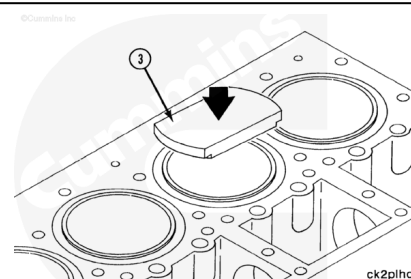
Remove the pistons and connecting rods. Refer to Procedure 001-054 in Section 1.

(</qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html>)

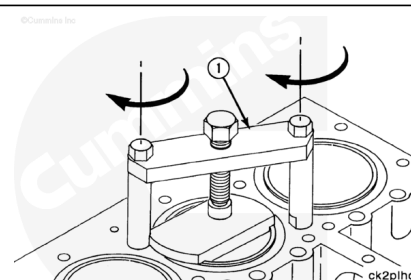


Use the cylinder liner clamping tool, Part Number 3824272, to seat the liner in the cylinder block.

Place the force plate (3) across the top of the liner with the step inside the liner bore.

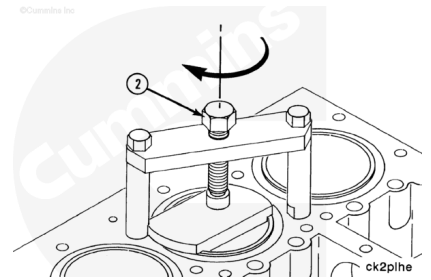


Place the installation tool bridge (1) across the liner and install the two cylinder head capscrews finger-tight to hold the bridge down. Rotate the force plate until the areas where the protrusion measurements will be taken are exposed.



⚠ CAUTION ⚠

Do not use an impact wrench to tighten the liner force plate screw. Excessive torque will cause damage to the liner installation tool.

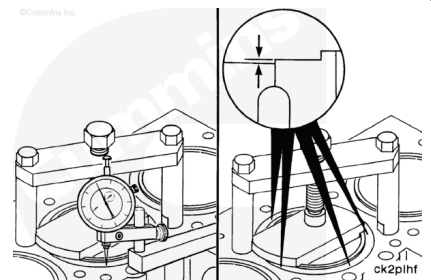


Tighten the forcing screw (2).

Torque Value: 136 n•m [100 ft-lb]

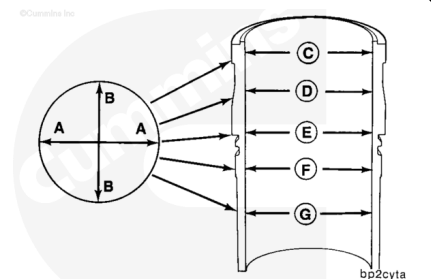
Use a depth gauge, Part Number 3164438, to measure the liner protrusion at four points that are 90 degrees apart.

Cylinder Liner Protrusion		
mm		in
0.00	MIN	0.0000
0.12	MAX	0.0047



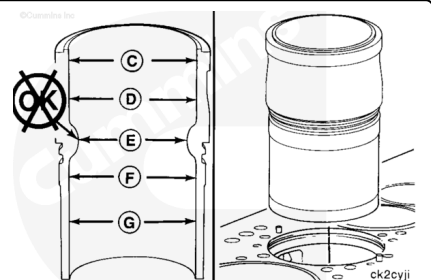
Maximum allowable variation in liner protrusion for each 180 degrees is 0.03mm [0.001 inch].

Measure the liner bore for out-of-roundness at points "C," "D," "E," "F," and "G." Measure each point in the direction "AA" and "BB". The bore **must not** be more than 0.10 mm [0.004 in] out-of-round.



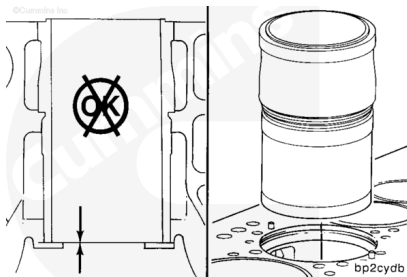
If the liner bore is more than 0.10 mm [0.004 in] out-of-round, remove the liner so the cylinder block liner bore can be measured.

The block counterbore diameters above and below the cylinder block counterbore area are **not** critical dimensions, and do **not** need to be measured.



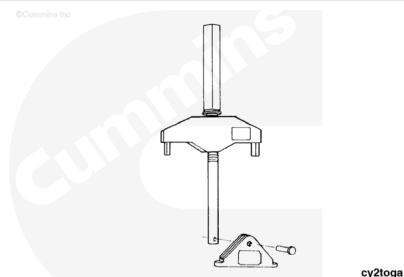
Use a feeler gauge to measure the liner-to-block clearance at the four block casting points.

If the clearance is less than 0.25 mm [0.010 in], replace the liner.



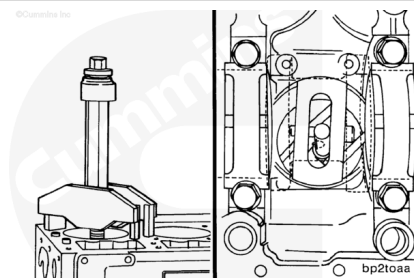
Remove

The cylinder liners can be removed by using the cylinder liner remover, Part Number 3163745, and cylinder liner remover plate, Part Number 3164979.



⚠ CAUTION ⚠

The liner puller must be installed and used as described to avoid damage to the cylinder block. The puller plate must be parallel to the main bearing saddles and must not overlap the liner outside diameter.

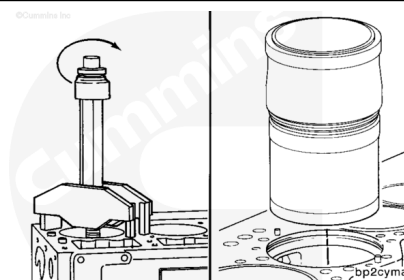


Insert the liner puller on the top of the cylinder block.

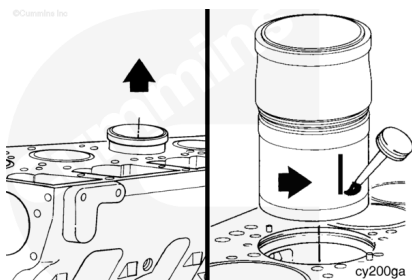
The liner puller **must** be centered on the top of the cylinder block.

Turn the puller jackscrew **clockwise** to loosen the liner from the cylinder block.

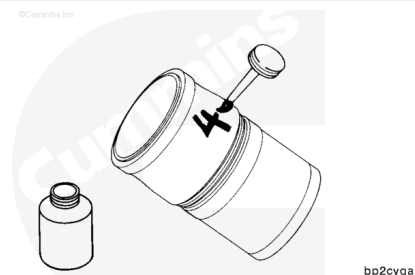
Use both hands to remove the liner.



When the liner is removed from the cylinder block, use Dykem®, or equivalent, to place a mark on the camshaft side of the liner to show liner orientation.



Use Dykem®, or equivalent, to mark the cylinder number on each liner.



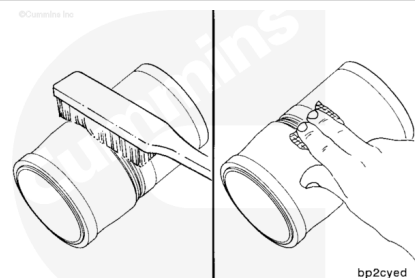
Clean and Inspect for Reuse

⚠ CAUTION ⚠

**Do not use any abrasives in the ring travel area of the liner.
The liner can be damaged.**

Use a soft wire brush to clean the flange seating area.

Use a fine fibrous abrasive pad such as Scotch-Brite™ 7448, or equivalent, to remove the remaining scale and rust.

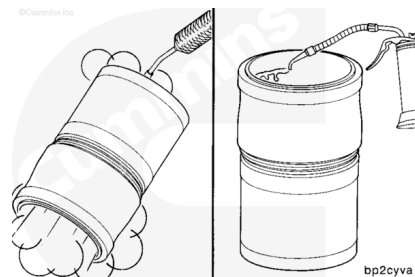


⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

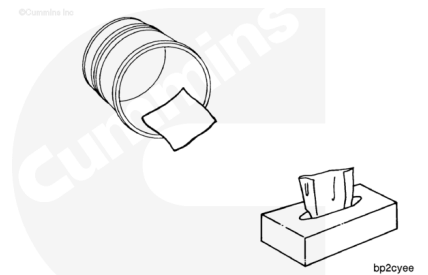
Use solvent or steam to clean the liners and dry with compressed air.

Use clean 15W-40 oil to lubricate the inside diameter of the liners.

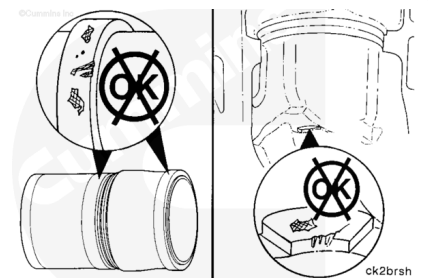
Allow the oil to soak into the liners for 5 to 10 minutes.

Use lint-free paper towels to wipe the excess oil from the inside of the liners.

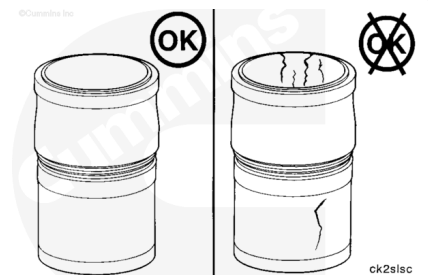
Continue to lubricate the inside of the liners and wipe them clean until the paper towel shows no gray or black residue.



Inspect the liners and cylinder block for dirt or damage.

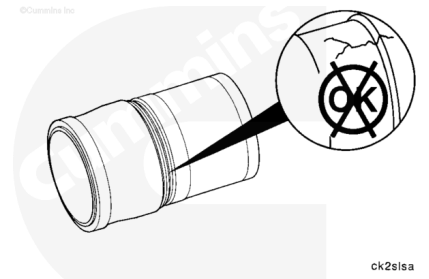


Inspect the liners for cracks on the inside and outside diameters.



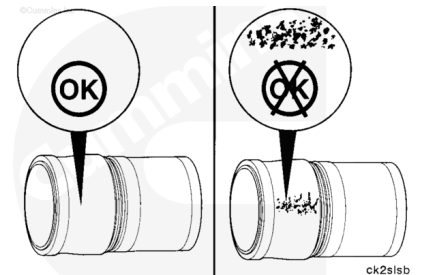
Inspect for cracks under the liner flange.

Cracks can also be detected using either magnetic inspection or the dye method.



Inspect the outside diameter for excessive corrosion or pitting. Liners with pitting generally can **not** be reused. However, if the pitting is light and can be removed with fine emery cloth, the liner can be reused.

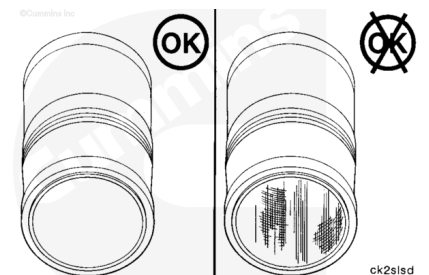
Pits **must not** be more than 1.60 mm [0.060 in] deep.



Inspect the inside diameter for vertical scratches deep enough to be felt with a fingernail.

If a fingernail catches in the scratch, the liner **must** be replaced.

Inspect the inside diameter for scuffing or scoring.

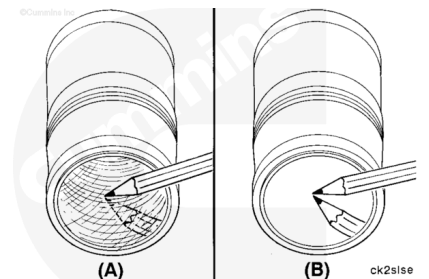


Inspect the inside diameter for liner bore polishing.

Note : M Series cylinder liners are not lubricated.

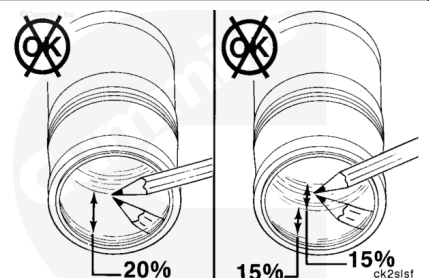
A moderate polish (A) produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

A heavy polish (B) produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.



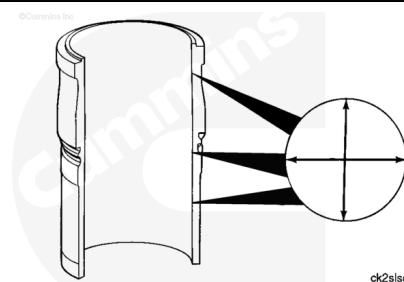
Replace the liner if:

- A heavy polish is present over 20 percent of the piston ring travel area.
- 30 percent of the piston ring travel area has both moderate and heavy polish and one-half (15 percent) is heavy polish.



Measure

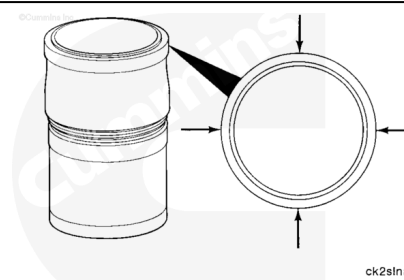
Use a dial bore gauge to measure the liner's inside diameter in four places, 90-degrees apart, at the top and bottom of the piston travel area.



Cylinder Liner Inside Diameter

mm		in
125.000	MIN	4.9213
125.095	MAX	4.9250

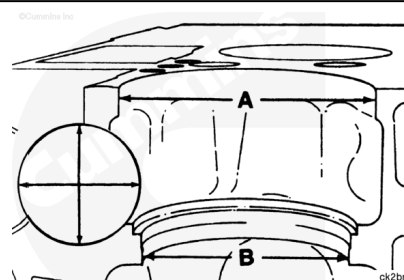
Measure the liner top press fit area outside diameter.



Cylinder Liner Top Press Fit Outside Diameter

mm		in
145.938	MIN	5.7456
145.976	MAX	5.7471

Measure the cylinder block upper liner bore (A).



Cylinder Block Upper Liner Bore Inside Diameter (A)

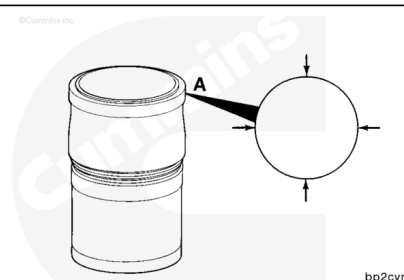
mm		in
145.900	MIN	5.7441
145.950	MAX	5.7460

Measure the cylinder block liner seal seat bore (B) 8.0 to 13.5 mm [0.32 to 0.53 in] below the counterbore.

Cylinder Block Liner Seal Seat Bore Inside Diameter (B)

mm		in
138.063	MIN	5.4355
138.113	MAX	5.4375

Measure the liner outside diameter (A).



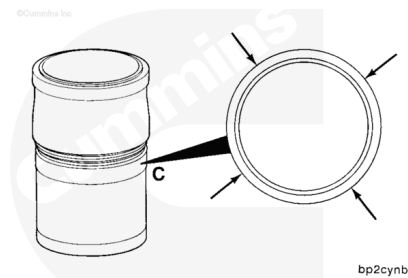
Cylinder Liner Top Press Fit Outside Diameter (A)

mm		in
145.938	MIN	5.7456
145.976	MAX	5.7471

The cylinder block liner counterbore flange diameter is **not** a critical dimension and does **not** need to be measured.

Measure the cylinder liner seal seat outside diameter (C).

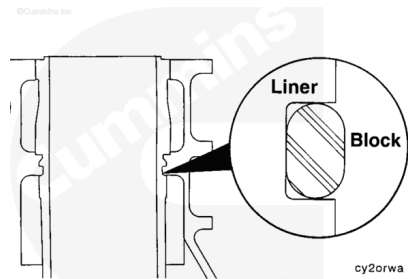
Cylinder Liner Seal Seat Outside Diameter (C)		
mm		in
137.937	MIN	5.4306
138.013	MAX	5.4336



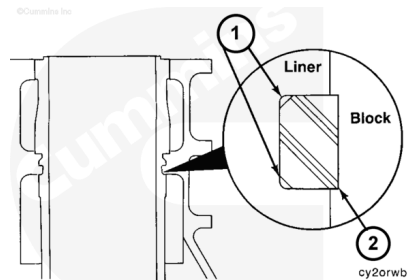
Install

The present o-ring has been produced by two distinctly different manufacturing processes. Following are those two methods and the installation procedures:

The molded o-ring has a symmetrical cross section. The edges are rounded. It does **not** require specific care in orientation of the o-ring other than normal prevention of rolling the o-ring.

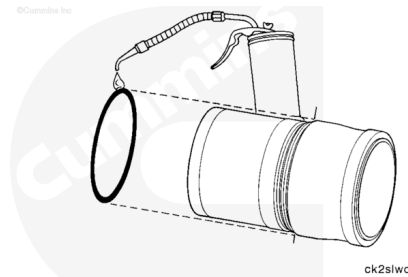


The lathe-cut o-ring has a straight outside with sharp corners. The inside has two 45-degree chamfers (1) so there are no sharp corners next to the liner. This o-ring **must** be installed with the straight side (2) facing the block and the chamfered side next to the liner. If the o-ring is **not** installed this way, liner bore distortion can occur.



⚠ CAUTION ⚠

The liners must be installed within 30 seconds after being lubricated with oil. If the liners are not installed within this time limit, the o-rings will swell and be damaged when the liner is installed.



Make sure the cylinder block and all parts are clean before assembly.

Use clean 15W-40 oil to coat the liner o-ring seals.

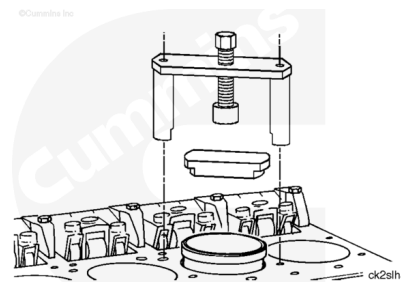
Install the o-ring seal on the liner.

Install the liner into the cylinder block.

When acceptable reused liners are installed, rotate the liners 90 degrees from their original positions in the engine. The thrust and anti-thrust surfaces **must** face the front and back of the cylinder block.

Use cylinder liner driver, Part Number 3824272, and a leather mallet to drive the liner into the cylinder block bore.

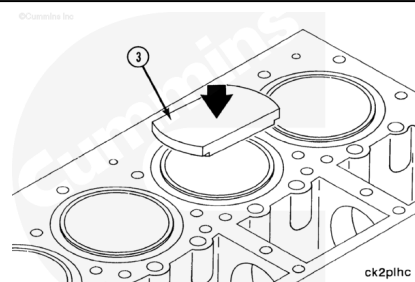
If the liner does **not** seat properly, remove the liner. Inspect the counterbore seat and liner for nicks, burrs, or dirt. Install the liner again.



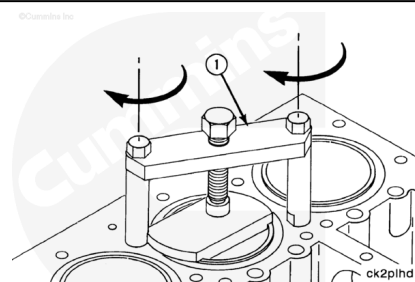
The cylinder liner protrusion **must** be checked before installing the pistons and connecting rods.

Use the cylinder liner clamping tool, Part Number 3824272, to seat the liner in the cylinder block.

Place the force plate (3) across the top of the liner with the step inside the liner bore.



Place the installation tool bridge (1) across the liner and install the two cylinder head capscrews finger-tight to hold the bridge down. Rotate the force plate until the areas where the protrusion measurements will be taken are exposed.

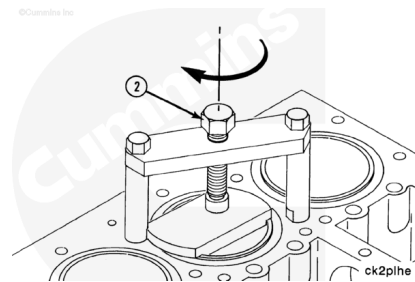


⚠ CAUTION ⚠

Do not use an impact wrench to tighten the liner force plate screw. Excessive torque will cause damage to the liner installation tool.

Tighten the forcing screw (2).

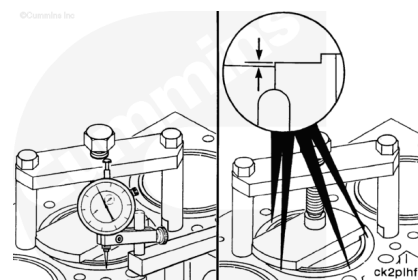
Torque Value: 136 n•m [100 ft-lb]



Use a depth gauge, Part Number 3164438, to measure the liner protrusion at four points that are 90 degrees apart.

Cylinder Liner Protrusion

mm		in
0.00	MIN	0.0000
0.12	MAX	0.0047

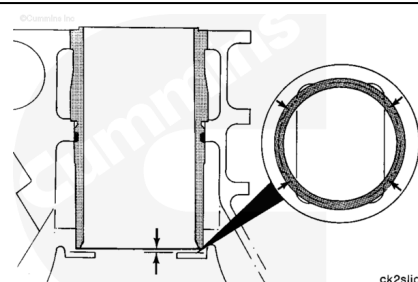


Maximum allowable variation in liner protrusion for each 180 degrees is 0.03mm [0.001 inch].

Use a feeler gauge to measure the liner-to-block clearance at the four-block casting points.

Cylinder Liner-to-Block Clearance

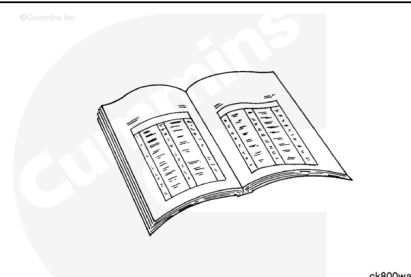
mm		in
0.25	MIN	0.010



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations

- Install the pistons and connecting rods. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html)
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/35/35-001-046-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-002-004-tr.html)
- Fill the cooling system with coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html)
- Operate the engine to normal operating temperature and check for leaks.